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Course Code: CSA1402

Compiler Design Lab Programs

18. Write a LEX program to count the number of Macros defined and header files included in the C program.

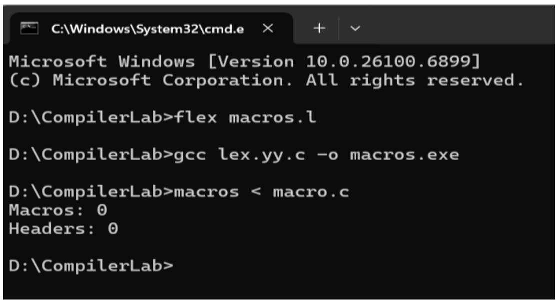
Code:

```
%{
#include <stdio.h>
int macros=0, headers=0;
%}
%%
^[ \t]*#define\b    { macros++; }
^[ \t]*#include\b   { headers++; }
.|\n                ;
%%
int main(){ yylex(); printf("Macros: %d\nHeaders: %d\n", macros, headers); return 0; }
int yywrap(){ return 1; }
```

Output:

```
#define PI 3.14
#include <stdio.h>
#include <conio.h>
makefile
Copy code
Macros: 1
Headers: 2
```

Output Screenshot:



19. Write a LEX program to print all HTML tags in the input file.

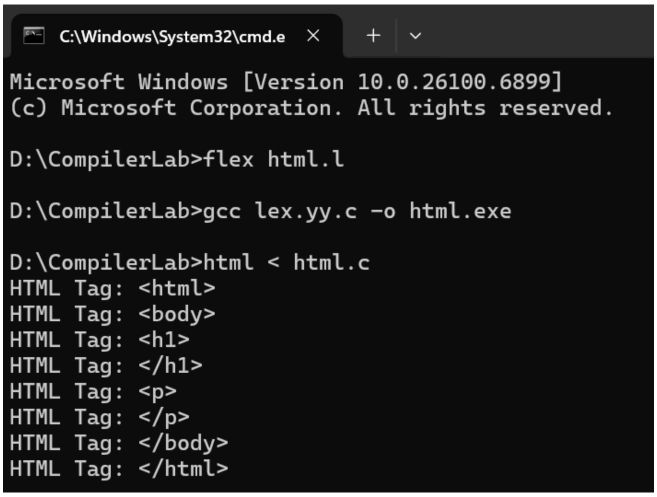
Code:

```
%{
#include <stdio.h>
%}
%%
\<[^>]+\>    { printf("HTML Tag: %s\n", yytext); }
.|\n         ;
%%
int main(){ yylex(); return 0; }
int yywrap(){ return 1; }
```

Output:

```
<html>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
HTML Tag: <html>
HTML Tag: <body>
HTML Tag: <h1>
HTML Tag: </h1>
HTML Tag: <p>
HTML Tag: </p>
HTML Tag: </body>
HTML Tag: </html>
```

Output Screenshot:



20.Write a LEX program which adds line numbers to the given C program file and display the same in the standard output.

Code:

```
%{
#include <stdio.h>
int lineno = 1;
%}
%%
[^\n]*\n   { printf("%4d: %s", lineno++, yytext); }
[^\n]+     { printf("%4d: %s\n", lineno++, yytext); }
%%
int main(){ yylex(); return 0; }
int yywrap(){ return 1; }
```

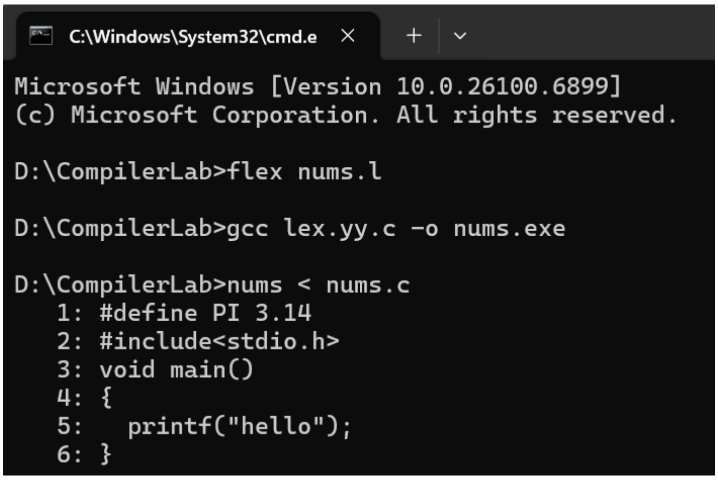
Output:

```
#define PI 3.14
#include<stdio.h>
void main()
{
  printf("hello");
}
cpp
```

Copy code

```
1: #define PI 3.14
2: #include<stdio.h>
3:
4: void main()
5: {
6:   printf("hello");
7: }
```

Output Screenshot:



21.Write a LEX specification count the number of characters, number of lines & number of words.

Code:

```
%{
#include <stdio.h>
int chars=0, words=0, lines=0;
%}
%%
[ \t]+      ;
\n          { lines++; chars++; }
[A-Za-z0-9_]+{ words++; chars += yyleng; }
.           { chars += yyleng; }
%%
```

```
int main(){ yylex(); printf("Chars:%d Words:%d Lines:%d\n", chars, words, lines); return 0;
}

int yywrap(){ return 1; }
```

Output:

Hello world

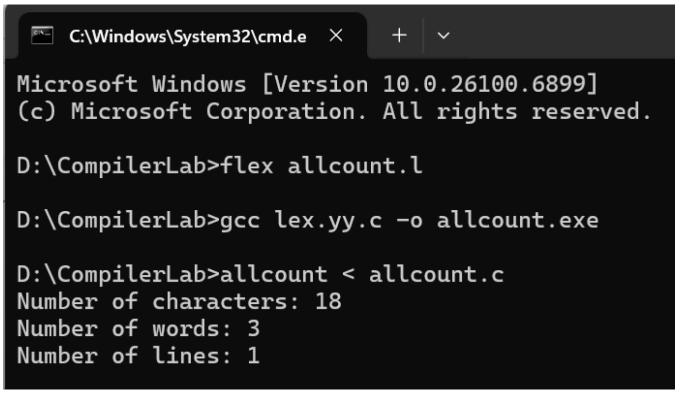
Welcome

Chars:19

Words:3

Lines:2

Output Screenshot:



22.Write a LEX program to count the number of comment lines in a given C program and eliminate them and write into another file.

Code:

```
%{

#include <stdio.h>

int comment_lines=0;

FILE *out;

%}

%%

"//".*              { comment_lines++; }

"/"*([^*]|\*+[^*/])*\*"/"    { /* attempt to match block comment */ comment_lines++; }

[^\n]+              { fprintf(out, "%s", yytext); }

\n                  { fprintf(out, "\n"); }

%%
```

```
int main(){
  yyin = stdin;
  out = fopen("output.c","w");
  if(!out){ perror("output.c"); return 1; }
  yylex();
  fclose(out);
  printf("Comment lines: %d\nNon-comment code written to output.c\n", comment_lines);
  return 0;
}
int yywrap(){ return 1; }
```

Output:

```
#include<stdio.h>
int main()
{
  int a; /*var declaration*/
  printf("hi");
  return 0;
}
Comment lines: 2
Non-comment code written to output.c
```

Output Screenshot: